EA-0018 - AQELYN Engineering Archive

AQELYN - EA-0018 Engineering Archive

IS-018 - AQELYN Automated Response & Orchestration Engine

Archive ID: EA-0018

Implementation Specification: IS-018

Component: AQELYN Automated Response & Orchestration Engine

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: COMPLETE

Repository Impact: No top-level repository structure changes

Breaking Changes: None

Engineering Phase: Phase 3

Predecessor Archives: EA-0001 through EA-0017

Next Specification: IS-019 - AQELYN Security Data Lake & Telemetry Platform

Document Control

Field	Value
Document	Engineering Archive EA-0018
Specification	IS-018 - AQELYN Automated Response & Orchestration Engine
Publication Format	Markdown, PDF, HTML, ZIP
Source of Truth	MD/EA-0018.md
Archive Rule	Implementation Specification -> Engineering Archive -> Continue
Repository Rule	Fixed repository structure; no redesign
Completion State	IS-018 complete; EA-0018 generated

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

Engineering Archive	Implementation Specification	Component
EA-0001	IS-001	AQELYN Kernel
EA-0002	IS-002	Universal Object Model
EA-0003	IS-003	AQELYN Event Bus

| EA-0004 | IS-004 | AQELYN Evidence Engine |

| EA-0005 | IS-005 | AQELYN Knowledge Graph |

| EA-0006 | IS-006 | AQELYN Trust Engine |

| EA-0007 | IS-007 | AQELYN Mission Engine |

| EA-0008 | IS-008 | AQELYN Workflow Engine |

| EA-0009 | IS-009 | AQELYN Policy Engine |

| EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |

| EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |

| EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |

| EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |

| EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |

| EA-0015 | IS-015 | AQELYN Security Operations (SOC) Engine |

| EA-0016 | IS-016 | AQELYN Digital Forensics Engine |

| EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |

| EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |

The fixed repository structure remains:

```
AQELYN/
■■■ archive/
■■■ blueprint/
■■■ docs/
■■■ src/
■■■ tests/
■■■ tools/
■■■ build/
■■■ releases/
■■■ scripts/
■■■ assets/
■■■ examples/
■■■ plugins/
■■■ sdk/
■■■ api/
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification
↓
Generate Engineering Archive
↓
Continue
```

No Engineering Archive may be skipped.

2. IS-018 Specification Identity

```
Specification ID: IS-018
Name: AQELYN Automated Response & Orchestration Engine
Engineering Archive Target: EA-0018
Project: AQELYN
System Type: Cyber Security Operating Environment
Status: Complete
Predecessor: IS-017 - AQELYN Threat Detection & Analytics Engine
```

3. Purpose

The AQELYN Automated Response & Orchestration Engine provides intelligent, policy-driven automation for coordinating detection, investigation, containment, remediation, recovery, and post-incident activities across the AQELYN platform.

It transforms detections and analyst decisions into repeatable, auditable response workflows.

It answers:

```
What response should be executed?
Can this incident be handled automatically?
Which playbook should be selected?
Which systems require containment?
Which approvals are required?
How should recovery proceed?
How can response actions be audited?
What lessons should feed future automation?
```

4. Mission

The engine shall provide:

```
Security orchestration
Automated response
Playbook execution
Workflow automation
Incident containment
Incident remediation
Recovery orchestration
Approval workflows
Human-in-the-loop decision support
Response auditing
Response metrics
Event publication
```

5. Scope

5.1 In Scope

```
Security playbooks
Incident response
Automated containment
Automated remediation
Recovery workflows
Approval management
Task orchestration
Cross-engine coordination
Response dashboards
Response reporting
```

5.2 Out of Scope

```
Operating system patch management
Enterprise IT service management
Business process automation
Physical access control
Industrial control systems
Device firmware updates
```

6. Dependencies

IS-018 depends on:

```
IS-001 AQELYN Kernel
IS-002 Universal Object Model
IS-003 AQELYN Event Bus
IS-004 AQELYN Evidence Engine
IS-005 AQELYN Knowledge Graph
IS-006 AQELYN Trust Engine
```

```
IS-007 AQELYN Mission Engine
IS-008 AQELYN Workflow Engine
IS-009 AQELYN Policy Engine
IS-010 AQELYN Compliance & Governance Engine
IS-011 Identity & Access Governance Engine
IS-012 Asset & Configuration Governance Engine
IS-013 Risk Intelligence Engine
IS-014 Threat Intelligence Fusion Engine
IS-015 Security Operations Engine
IS-016 Digital Forensics Engine
IS-017 Threat Detection & Analytics Engine
```

7. High-Level Architecture

```
AQELYN Automated Response & Orchestration Engine
■
■■■ Response Engine
■■■ Orchestration Engine
■■■ Playbook Engine
■■■ Automation Engine
■■■ Approval Engine
■■■ Containment Engine
■■■ Recovery Engine
■■■ Metrics Engine
■■■ Dashboard Service
■■■ Workflow Connector
■■■ Policy Connector
■■■ Knowledge Graph Connector
■■■ Event Publisher
```

8. Functional Requirements

FR-018-001 - Playbook Execution

The engine shall execute:

```
Incident playbooks
Containment playbooks
Recovery playbooks
Investigation playbooks
Compliance playbooks
Custom response playbooks
```

FR-018-002 - Automated Containment

Support automated:

```
Host isolation
Account suspension
Credential revocation
Network segmentation
Firewall updates
Application quarantine
```

FR-018-003 - Automated Remediation

Provide:

```
Configuration rollback
Policy enforcement
Threat removal
Credential reset
Service restart
Asset restoration
```

FR-018-004 - Approval Workflows

Support:

```
Manual approval
Multi-level approval
Emergency override
Role-based authorization
Time-limited approvals
Approval auditing
```

FR-018-005 - Workflow Orchestration

Coordinate:

```
Detection
Investigation
Containment
Remediation
Recovery
Post-incident review
```

FR-018-006 - Metrics & Reporting

Generate:

```
Response metrics
Playbook metrics
Automation success rates
Recovery statistics
Executive reports
Operational dashboards
```

FR-018-007 - Event Publication

Publish standardized events:

```
response.started
response.completed
playbook.executed
containment.completed
recovery.completed
approval.granted
```

9. Non-Functional Requirements

The engine shall provide:

```
High availability
Scalable orchestration
Policy-driven automation
Full auditability
Low-latency execution
Repository stability
Backward compatibility
Continuous operation
```

10. Core Response Workflow

```
Threat Detected
↓
Policy Evaluation
↓
Playbook Selection
```

```
↓
Approval (if required)
↓
Automated Response
↓
Containment
↓
Remediation
↓
Recovery
↓
Metrics & Audit
↓
Response Published
```

11. Internal Component Architecture

The AQELYN Automated Response & Orchestration Engine is implemented as a modular orchestration subsystem integrated with the AQELYN Kernel, Event Bus, Workflow Engine, Policy Engine, Security Operations Engine, and Threat Detection Engine.

```
AQELYN Automated Response & Orchestration Engine
■
■■■ Response Engine
■■■ Orchestration Engine
■■■ Playbook Engine
■■■ Automation Engine
■■■ Approval Engine
■■■ Containment Engine
■■■ Remediation Engine
■■■ Recovery Engine
■■■ Metrics Engine
■■■ Dashboard Service
■■■ Workflow Connector
■■■ Policy Connector
■■■ Knowledge Graph Connector
■■■ Evidence Connector
■■■ Event Publisher
```

12. Component Specifications

12.1 Response Engine

Coordinates complete incident response execution.

Capabilities:

```
Response initiation
Task coordination
Execution tracking
Completion validation
Failure handling
```

12.2 Orchestration Engine

Coordinates actions across AQELYN subsystems.

Functions:

```
Cross-engine orchestration
Task sequencing
Dependency management
Parallel execution
Workflow synchronization
```

12.3 Playbook Engine

Executes predefined and custom playbooks.

Supported playbooks:

```
Containment
Investigation
Recovery
Compliance
Threat response
Custom enterprise workflows
```

12.4 Automation Engine

Executes automated operational actions.

Supports:

```
Host isolation
Credential reset
Firewall updates
Service restart
Application quarantine
Asset restoration
```

12.5 Approval Engine

Manages response authorization.

Supports:

```
Role approval
Multi-stage approval
Emergency approval
Delegated approval
Approval auditing
```

12.6 Containment Engine

Coordinates containment activities.

Capabilities:

```
Endpoint isolation
Network isolation
Account suspension
Application blocking
Threat containment
```

12.7 Remediation Engine

Coordinates corrective actions required to eliminate or reduce a threat condition.

Capabilities:

```
Configuration rollback
Threat removal
Policy remediation
Credential reset
Vulnerability remediation handoff
System hardening
```

12.8 Recovery Engine

Coordinates recovery activities.

Supports:

```
Asset recovery
Configuration restoration
Credential reactivation
Service recovery
Validation testing
```

12.9 Metrics Engine

Calculates operational metrics.

Provides:

```
Mean response time
Automation rate
Recovery duration
Playbook efficiency
Success rate
```

12.10 Dashboard Service

Provides operational dashboards.

Displays:

```
Active responses
Running playbooks
Pending approvals
Containment status
Recovery status
Automation metrics
```

13. Universal Object Model Extensions

13.1 ResponseAction

```
ResponseAction:
response_id
playbook
status
owner
timestamp
```

13.2 Playbook

```
Playbook:
playbook_id
version
workflow
approvals
```

13.3 Approval

```
Approval:
approval_id
approver
decision
timestamp
```

13.4 RecoveryOperation

```
RecoveryOperation:
recovery_id
asset
status
completion
```

14. Knowledge Graph Integration

Relationships:

```
Incident
↓
triggers
↓
Response

Response
↓
executes
↓
Playbook

Playbook
↓
contains
↓
Task

Task
↓
affects
↓
Asset

Recovery
↓
restores
↓
Mission
```

15. Event Bus Integration

15.1 Response Events

```
response.started
response.completed
response.failed
```

15.2 Playbook Events

```
playbook.executed
playbook.failed
playbook.completed
```

15.3 Approval Events

```
approval.requested
approval.granted
approval.denied
```

15.4 Recovery Events

```
recovery.started
recovery.completed
```

```
recovery.failed
```

16. Workflow Engine Integration

Consumes:

```
Workflow definitions
Execution state
Task dependencies
Workflow history
```

17. Policy Engine Integration

Consumes:

```
Response policies
Approval rules
Containment rules
Automation policies
```

18. Security Operations Integration

Supports:

```
Incident response
SOC escalation
Threat hunting
Operational coordination
```

19. Threat Detection Integration

Consumes:

```
Threat detections
Threat scores
Behavior analytics
Predictions
```

20. Digital Forensics Integration

Consumes:

```
Evidence
Timeline reports
Artifact verification
Investigation findings
```

21. Compliance Integration

Supports:

```
Audit records
Approval evidence
Response evidence
Regulatory reporting
```

22. Public APIs

22.1 Response API

```
GET /responses
POST /responses
GET /responses/{id}
```

22.2 Playbook API

```
GET /playbooks
POST /playbooks
```

22.3 Approval API

```
GET /approvals
POST /approvals
```

22.4 Recovery API

```
GET /recovery
POST /recovery
```

23. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
■■■ src/
■ ■■■ response_orchestration/
■■■ tests/
■ ■■■ response_orchestration/
■■■ docs/
■ ■■■ response_orchestration/
■■■ api/
■ ■■■ response_orchestration/
■■■ archive/
```

No top-level repository modifications are permitted.

24. Security Architecture

The AQELYN Automated Response & Orchestration Engine is the execution subsystem responsible for securely coordinating automated response actions across the AQELYN platform.

Every response shall be:

```
Policy-governed
Role-authorized
Evidence-backed
Fully auditable
Traceable
Reproducible
Mission-aware
Risk-aware
```

24.1 Security Principles

```
Zero Trust
Least Privilege
Defense in Depth
Policy Enforcement
Human-in-the-Loop
Secure Automation
Continuous Verification
Separation of Duties
```

24.2 Authorization Model

Supported operational roles:

```
SOC Analyst
Incident Commander
Automation Operator
Mission Owner
Compliance Officer
Security Administrator
Playbook Author
Automation Service
```

All privileged automation shall be authorized through the AQELYN Policy Engine.

24.3 Automation Integrity

Automation records shall maintain:

```
Unique response identifier
Playbook version
Execution history
Actor or service identity
Approval references
Evidence references
Policy decision references
Outcome state
Audit trail
```

Automation and response history shall be append-only.

24.4 Approval Integrity

Approval records shall maintain:

```
Approver identity
Decision
Timestamp
Scope
Expiration
Reason
Policy reference
Evidence reference
```

Approvals shall not be destructively modified after decision recording.

25. Response Lifecycle

25.1 Incident Response Lifecycle

```
Threat Detected
↓
Policy Evaluation
```

```
↓
Playbook Selected
↓
Approval (if required)
↓
Automated Response
↓
Containment
↓
Remediation
↓
Recovery
↓
Closed
```

25.2 Playbook Lifecycle

```
Draft
↓
Validated
↓
Approved
↓
Published
↓
Executed
↓
Archived
```

25.3 Approval Lifecycle

```
Requested
↓
Assigned
↓
Reviewed
↓
Approved / Denied
↓
Recorded
```

25.4 Recovery Lifecycle

```
Recovery Started
↓
Validation
↓
Service Restoration
↓
Verification
↓
Completed
```

26. Continuous Automation

The engine continuously evaluates:

```
Running playbooks
Policy compliance
Approval status
Automation health
Execution failures
Recovery progress
Mission impact
Operational risk
```

27. Performance Requirements

The engine shall support:

```
Low-latency orchestration
Parallel playbook execution
Enterprise-scale automation
Concurrent approvals
Large workflow execution
Continuous response processing
```

28. Scalability Requirements

The engine shall scale to support:

```
Thousands of concurrent incidents
Millions of workflow events
Large enterprise environments
Hybrid deployments
Distributed automation services
Global operations
```

29. Audit Requirements

Every orchestration activity shall generate immutable audit records.

Audit events include:

```
Response initiated
Playbook executed
Approval granted
Containment completed
Recovery completed
Policy decision
Automation failure
```

30. Failure Handling

30.1 Playbook Failure

```
Execution halted
Failure recorded
Rollback initiated
```

30.2 Automation Failure

```
Retry scheduled
Analyst notified
Audit generated
```

30.3 Approval Failure

```
Execution paused
Escalation initiated
Decision recorded
```

30.4 Recovery Failure

```
Recovery halted
Validation repeated
Incident escalated
```

31. Testing Strategy

31.1 Unit Testing

Validate:

```
Response Engine
Playbook Engine
Automation Engine
Approval Engine
Recovery Engine
Metrics Engine
```

31.2 Integration Testing

Verify interaction with:

```
Kernel
Workflow Engine
Policy Engine
SOC Engine
Threat Detection Engine
Digital Forensics Engine
Evidence Engine
Knowledge Graph
Compliance Engine
```

31.3 System Testing

Validate:

```
Playbook execution
Incident orchestration
Automated containment
Automated remediation
Recovery workflows
Operational dashboards
```

31.4 Security Testing

Verify:

```
Authorization
Policy enforcement
Approval workflows
Audit logging
Execution integrity
```

31.5 Regression Testing

Verify IS-001 through IS-017 remain unaffected.

32. Acceptance Criteria

IS-018 is complete when:

```
Response Engine implemented
Playbook Engine implemented
Automation Engine implemented
Approval Engine implemented
Recovery Engine implemented
```

```
Metrics Engine implemented
Repository unchanged
Testing documented
```

33. Repository Validation

Repository structure remains unchanged.

```
AQELYN/
■■■ src/response_orchestration/
■■■ tests/response_orchestration/
■■■ docs/response_orchestration/
■■■ api/response_orchestration/
■■■ archive/
```

No top-level repository modifications are permitted.

34. Engineering Summary

IS-018 introduces the AQELYN Automated Response & Orchestration Engine, providing enterprise-grade orchestration of incident response, playbook execution, containment, remediation, recovery, approval workflows, and response metrics.

Major capabilities include:

```
Security Orchestration
Automated Response
Playbook Execution
Workflow Coordination
Incident Containment
Automated Remediation
Recovery Management
Approval Workflows
Operational Dashboards
Response Metrics
Policy-driven Automation
```

The engine integrates with the Workflow Engine, Policy Engine, Threat Detection & Analytics Engine, Security Operations Engine, Digital Forensics Engine, Evidence Engine, and Knowledge Graph while preserving repository stability and backward compatibility.

35. Specification Status

```
Specification ID : IS-018
Title : AQELYN Automated Response & Orchestration Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0018
```

Engineering workflow status:

```
EA-0001 COMPLETE
EA-0002 COMPLETE
EA-0003 COMPLETE
EA-0004 COMPLETE
EA-0005 COMPLETE
EA-0006 COMPLETE
EA-0007 COMPLETE
EA-0008 COMPLETE
EA-0009 COMPLETE
EA-0010 COMPLETE
EA-0011 COMPLETE
EA-0012 COMPLETE
```

```
EA-0013 COMPLETE
EA-0014 COMPLETE
EA-0015 COMPLETE
EA-0016 COMPLETE
EA-0017 COMPLETE
IS-018 COMPLETE
EA-0018 READY FOR GENERATION
```

36. EA-0018 Engineering Objective

The objective of IS-018 was to introduce a dedicated Automated Response & Orchestration Engine that enables AQELYN to select, approve, execute, audit, and validate automated response playbooks across incident response, containment, remediation, and recovery workflows.

The engine extends AQELYN from detection and security operations into controlled, policy-driven response execution.

37. EA-0018 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

```
Response coordination
Cross-engine orchestration
Playbook execution
Automation action execution
Approval workflows
Containment operations
Remediation operations
Recovery operations
Response metrics
Dashboards
Workflow integration
Policy integration
Knowledge Graph integration
Evidence integration
Event publishing
```

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

38. Major Engineering Decisions

38.1 Decision 1 - Dedicated Automated Response & Orchestration Engine

Response and orchestration responsibilities are implemented as a standalone engine rather than embedded in SOC, Workflow, or Threat Detection.

Rationale:

```
Clear separation of operational analysis from automated response execution.
Independent lifecycle and scaling.
Better support for approval workflows and human-in-the-loop control.
Improved auditability of automated action.
```

38.2 Decision 2 - Human-in-the-Loop Automation

Automation is governed by policy and may require explicit approval before execution.

Benefits:

```
High-risk actions remain controlled.
Emergency overrides can be audited.
Approvals become evidence-backed.
Mission owners and incident commanders can participate in decisions.
```

38.3 Decision 3 - Playbooks as Governed Objects

Playbooks are versioned and tracked as Universal Object Model extensions.

Benefits:

```
Playbook execution is reproducible.
Changes can be audited.
Response outcomes can be linked to playbook versions.
Automation metrics can evaluate playbook effectiveness.
```

38.4 Decision 4 - Event-Driven Response Pipeline

Response, playbook, approval, containment, remediation, and recovery events are published through the AQELYN Event Bus.

Examples include:

```
response.started
response.completed
response.failed
playbook.executed
playbook.failed
approval.requested
approval.granted
approval.denied
recovery.started
recovery.completed
```

This maintains loose coupling between AQELYN engines.

38.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
ResponseAction
Playbook
Approval
RecoveryOperation
```

These extend the Universal Object Model without modifying existing object definitions.

39. Architectural Integration Summary

Engine	Integration
IS-001 Kernel	Runtime lifecycle and service registration
IS-002 Universal Object Model	Response, playbook, approval, recovery objects
IS-003 Event Bus	Response, playbook, approval, recovery events
IS-004 Evidence Engine	Response evidence, approval evidence, execution records
IS-005 Knowledge Graph	Incident, response, playbook, task, asset, recovery relationships
IS-006 Trust Engine	Evidence trust and automation confidence
IS-007 Mission Engine	Mission impact and recovery prioritization
IS-008 Workflow Engine	Workflow definitions, task dependencies, workflow history

| IS-009 Policy Engine | Response policies, approval rules, containment rules |

| IS-010 Compliance Engine | Audit records, approval evidence, regulatory reporting |

| IS-011 Identity Governance Engine | Account suspension, credential revocation, role authorization |

| IS-012 Asset Governance Engine | Host isolation, restoration, ownership, criticality |

| IS-013 Risk Intelligence Engine | Operational risk and risk-aware response decisions |

| IS-014 Threat Intelligence Engine | Threat context for response selection |

| IS-015 SOC Engine | Incident response, escalation, operational coordination |

| IS-016 Digital Forensics Engine | Evidence, timeline reports, investigation findings |

| IS-017 Threat Detection Engine | Detections, scores, analytics, predictions |

No existing engine required redesign.

40. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/response_orchestration/
■■■ tests/response_orchestration/
■■■ api/response_orchestration/
■■■ docs/response_orchestration/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

41. Security Impact Summary

The specification introduces response-orchestration-specific security controls:

```
Policy-driven automation
Role-authorized response execution
Human-in-the-loop approval
Immutable response audit trail
Playbook version control
Approval integrity
Execution evidence linkage
Rollback and failure handling
Mission-aware response control
Risk-aware containment decisions
```

No reduction in the security posture of existing components was identified.

42. Capabilities Added

The engine enables AQELYN to support:

```
Security orchestration
Automated response
Playbook execution
Incident containment
Automated remediation
Recovery orchestration
Approval workflows
Task orchestration
Cross-engine coordination
```

```
Response dashboards
Response metrics
Response reporting
```

43. Risks Identified

Risk	Mitigation
Unsafe automated response	Policy enforcement and human-in-the-loop approvals
Incorrect playbook selection	Policy rules, risk context, and SOC oversight
Automation failure	Retry, rollback, escalation, audit
Approval bottlenecks	Emergency override and delegated approval
Recovery failure	Validation testing and incident escalation
Excessive automation privileges	Least privilege and role authorization
Playbook drift	Versioning, validation, approval lifecycle
Poor auditability	Immutable response and approval records

No critical architectural risks were identified that require redesign.

44. Verification Summary

The specification defines verification for:

```
Unit testing
Integration testing
System testing
Security testing
Regression testing
```

Acceptance criteria cover response engine, playbook engine, automation engine, approval engine, recovery engine, metrics engine, repository validation, and testing documentation.

45. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

```
Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline
```

46. Dependencies

Required:

```
EA-0001 through EA-0017
IS-001 through IS-017
```

Enables:

```
IS-019 and subsequent telemetry, data lake, automation, operations, and platform-scale components
```

47. Completion Record

```
Engineering Archive : EA-0018
Implementation Specification : IS-018
Title : AQELYN Automated Response & Orchestration Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue
```

48. Archive Index Update

```
EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
EA-0013 IS-013 AQELYN Risk Intelligence Engine
EA-0014 IS-014 AQELYN Threat Intelligence Fusion Engine
EA-0015 IS-015 AQELYN Security Operations (SOC) Engine
EA-0016 IS-016 AQELYN Digital Forensics Engine
EA-0017 IS-017 AQELYN Threat Detection & Analytics Engine
EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
```

49. Engineering Phase Status

```
Completed Engineering Archives : EA-0001 through EA-0018

Current Status:
EA-0018 COMPLETE

Next Implementation Specification:
IS-019 - AQELYN Security Data Lake & Telemetry Platform
```

EA-0018 is completed and archived. The engineering workflow is consistent with the project rule:

```
Implementation Specification -> Engineering Archive -> Continue
```

From this point onward, the next engineering artifact is IS-019.

50. Engineering Archive Publication Standard

EA-0018 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
■
■■■ diagrams/
■ ■■■ Architecture.svg
■ ■■■ Component.svg
■ ■■■ Workflow.svg
■ ■■■ EventFlow.svg
■ ■■■ Integration.svg
■
■■■ examples/
■ ■■■ example_*.md
■
■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
■
■■■ images/
■
■■■ journal/
■ ■■■ Engineering_Journal.md
■
■■■ manifest/
■ ■■■ manifest.json
■
■■■ MD/
■ ■■■ EA-xxxx.md
■
■■■ PDF/
■ ■■■ EA-xxxx.pdf
■
■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

51. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-018-001	Execute playbooks	Sections 8, 12, 25	Complete
FR-018-002	Support automated containment	Sections 8, 12	Complete
FR-018-003	Support automated remediation	Sections 8, 12	Complete
FR-018-004	Support approval workflows	Sections 8, 12, 24, 25	Complete
FR-018-005	Coordinate workflow orchestration	Sections 8, 12, 16	Complete
FR-018-006	Generate metrics and reporting	Sections 8, 12	Complete
FR-018-007	Publish response events	Sections 8, 15, 38	Complete
NFR-018-001	High availability	Sections 9, 28	Complete
NFR-018-002	Scalable orchestration	Sections 9, 27, 28	Complete
NFR-018-003	Policy-driven automation	Sections 9, 17, 24	Complete
NFR-018-004	Full auditability	Sections 9, 29, 41	Complete
NFR-018-005	Low-latency execution	Sections 9, 27	Complete
NFR-018-006	Repository stability	Sections 23, 33, 40	Complete

52. Traceability Matrix

Source	Target	Relationship
IS-018 Purpose	EA-0018 Objective	Defines why the engine exists
Response Engine	FR-018 response coordination	Coordinates response execution
Orchestration Engine	FR-018-005	Coordinates cross-engine tasks
Playbook Engine	FR-018-001	Executes governed playbooks
Automation Engine	FR-018-002, FR-018-003	Executes containment and remediation actions
Approval Engine	FR-018-004	Manages approvals
Containment Engine	FR-018-002	Implements containment
Remediation Engine	FR-018-003	Implements remediation
Recovery Engine	Recovery lifecycle	Restores services and assets
Metrics Engine	FR-018-006	Calculates response metrics
Event Publisher	FR-018-007	Publishes response events
Workflow Engine Integration	Orchestration	Supplies workflow definitions and history
Policy Engine Integration	Authorization and rules	Supplies response and approval policies
SOC Integration	IS-015	Supplies incident response context
Threat Detection Integration	IS-017	Supplies detections and scores
Digital Forensics Integration	IS-016	Supplies evidence and timelines
Repository Validation	Repository Standard	Confirms no top-level redesign

53. Engineering Journal

Journal Entry - EA-0018

EA-0018 was created to archive completion of IS-018 - AQELYN Automated Response & Orchestration Engine.

The archive records the expansion of AQELYN into automated response and orchestration. IS-018 defines the structure needed to select playbooks, coordinate response, execute automation, manage approvals, contain incidents, perform remediation, orchestrate recovery, calculate metrics, and publish response events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Automated response must be modeled separately from SOC operations and Workflow execution. SOC owns operational incident handling, Workflow owns generic workflow mechanics, and the Automated Response & Orchestration Engine owns response selection, automation execution, approval enforcement, containment, remediation, recovery, and response metrics.

Governance Note

EA-0018 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

54. Examples

54.1 Example Response Action


```
response_id: RESP-0001
playbook: PB-CONTAIN-HOST-v1
status: in_progress
owner: incident_commander_01
timestamp: 2026-07-07T12:00:00Z
incident: INC-1001
```


54.2 Example Playbook


```
playbook_id: PB-CONTAIN-HOST
version: v1
workflow:
- validate_incident
- request_approval
- isolate_host
- collect_evidence
- notify_mission_owner
approvals:
- incident_commander
- mission_owner
```


54.3 Example Approval


```
approval_id: APR-2001
approver: mission_owner_01
decision: approved
timestamp: 2026-07-07T12:05:00Z
scope: isolate_asset_ASSET-0002
```


54.4 Example Response Event


```
{
"event_type": "response.started",
"response_id": "RESP-0001",
"playbook_id": "PB-CONTAIN-HOST",
"incident_id": "INC-1001",
"source_engine": "aqelyn_automated_response_orchestration_engine"
}
```


55. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0018.md
PDF/EA-0018.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_response_playbook.md
```

56. Final Archive Statement

EA-0018 is the Engineering Archive for IS-018 - AQELYN Automated Response & Orchestration Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0018 COMPLETE
IS-018 COMPLETE
NEXT: IS-019
```